

38. . Implement a context-aware chatbot using LangChain, retrieval, and an LLM.

Code:

```
from sentence_transformers import SentenceTransformer
from transformers import pipeline
import faiss
import numpy as np

# =====
# CONTEXT-AWARE AI CHATBOT
# =====

print("=====")
print("    CONTEXT-AWARE AI CHATBOT")
print("=====")

# =====
# 1. KNOWLEDGE BASE
# =====

documents = [
    "Artificial Intelligence is a branch of computer science "
    "that creates intelligent systems.",

    "Machine Learning is a subset of Artificial Intelligence "
    "that allows computers to learn from data.",

    "Deep Learning is a type of Machine Learning that uses "
```

"neural networks with multiple layers.",

"Natural Language Processing helps computers understand "

"and process human language.",

"Computer Vision enables computers to understand images "

"and videos.",

"Reinforcement Learning allows an agent to learn by "

"receiving rewards and penalties."

]

```
print("\nKnowledge base loaded.")
```

```
print("Total documents:", len(documents))
```

```
# =====
```

```
# 2. LOAD EMBEDDING MODEL
```

```
# =====
```

```
print("\nLoading embedding model...")
```

```
embedding_model = SentenceTransformer(
```

```
    "all-MiniLM-L6-v2"
```

```
)
```

```
print("Embedding model loaded.")
```

```
# =====  
# 3. CREATE DOCUMENT EMBEDDINGS  
# =====
```

```
print("\nCreating document embeddings...")
```

```
document_embeddings = embedding_model.encode(  
    documents  
)
```

```
document_embeddings = np.array(  
    document_embeddings  
)  
.astype("float32")
```

```
print("Embeddings created.")
```

```
# =====  
# 4. CREATE FAISS VECTOR DATABASE  
# =====
```

```
dimension = document_embeddings.shape[1]
```

```
index = faiss.IndexFlatL2(dimension)
```

```
index.add(document_embeddings)
```

```
print("FAISS vector database created.")
```

```
print("Vectors stored:", index.ntotal)
```

```
# =====
```

```
# 5. LOAD LLM
```

```
# =====
```

```
print("\nLoading language model...")
```

```
llm = pipeline(  
    "text2text-generation",  
    model="google/flan-t5-small"  
)
```

```
print("Language model loaded.")
```

```
# =====
```

```
# 6. CONVERSATION HISTORY
```

```
# =====
```

```
conversation_history = []
```

```
# =====
```

```
# 7. CHATBOT START
```

```
# =====
```

```
print("\n=====")
```

```
print("      CHATBOT READY")
print("=====")
```

```
print("Ask questions about Artificial Intelligence.")
print("The chatbot remembers the previous conversation.")
print("Type 'exit' to stop.")
```

```
while True:
```

```
    # -----
    # Get user question
    # -----
```

```
    question = input("\nYou: ")
```

```
    # -----
    # Exit
    # -----
```

```
    if question.lower() == "exit":
```

```
        print("\nChatbot: Goodbye!")
        break
```

```
    # =====
    # 8. CREATE QUERY FOR RETRIEVAL
```

```
# =====
```

```
# Use current question for document search
```

```
query_embedding = embedding_model.encode(  
    [question]  
)
```

```
query_embedding = np.array(  
    query_embedding  
) .astype("float32")
```

```
# =====
```

```
# 9. RETRIEVE RELEVANT DOCUMENTS
```

```
# =====
```

```
k = 2
```

```
distances, indices = index.search(  
    query_embedding,  
    k  
)
```

```
context = ""
```

```
for i in indices[0]:
```

```
context += documents[i] + "\n"
```

```
# =====
```

```
# 10. ADD PREVIOUS CONVERSATION
```

```
# =====
```

```
previous_context = ""
```

```
for user_question, bot_answer in conversation_history:
```

```
    previous_context += (  
        "User: " + user_question + "\n"  
    )
```

```
    previous_context += (  
        "Chatbot: " + bot_answer + "\n"  
    )
```

```
# =====
```

```
# 11. CREATE CONTEXT-AWARE PROMPT
```

```
# =====
```

```
prompt = ""
```

You are a helpful AI chatbot.

Use the knowledge context and previous conversation
to answer the user's current question.

Knowledge Context:

```
{}
```

Previous Conversation:

```
{}
```

Current Question:

```
{}
```

Give a short and clear answer.

```
"".format(
    context,
    previous_context,
    question
)

# =====
# 12. GENERATE ANSWER
# =====

result = llm(
    prompt,
    max_new_tokens=100
)

answer = result[0]["generated_text"]
```



```
# =====
```

```
# 13. SAVE CONVERSATION
```

```
# =====
```

```
conversation_history.append(  
    (question, answer)  
)
```

```
# =====
```

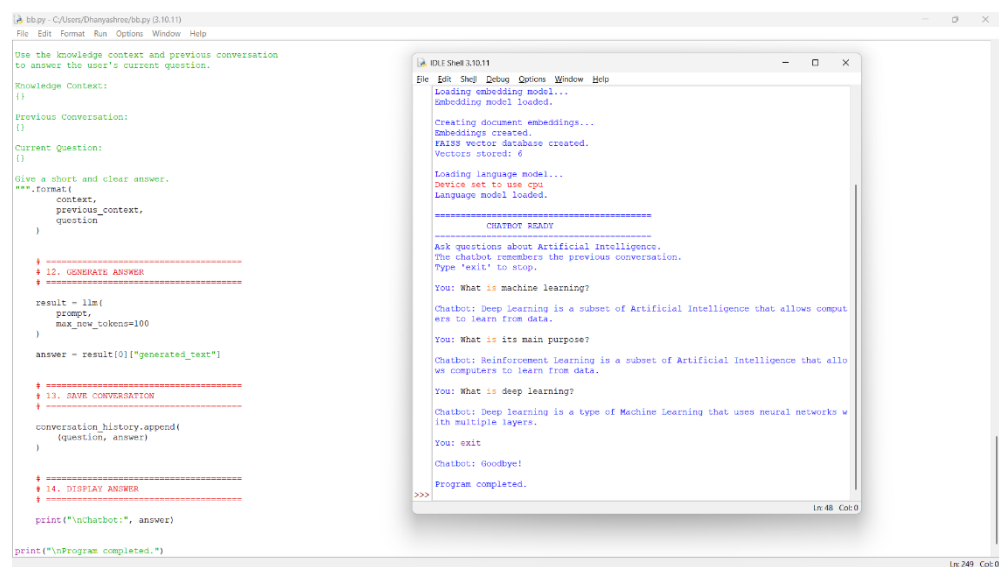
```
# 14. DISPLAY ANSWER
```

```
# =====
```

```
print("\nChatbot:", answer)
```

```
print("\nProgram completed.")
```

output:



The screenshot shows a Python IDE with two windows. The left window displays the source code for a chatbot, and the right window shows the output of the program.

Source Code (Left Window):

```
Use the knowledge context and previous conversation  
to answer the user's current question.  
Knowledge Context:  
{  
}  
Previous Conversation:  
{  
}  
Current Question:  
{  
}  
Give a short and clear answer.  
***.format(  
    context,  
    previous_context,  
    question  
)  
  
# 12. GENERATE ANSWER  
# =====  
result = llm(  
    prompt,  
    max_new_tokens=100  
)  
answer = result[0]["generated_text"]  
  
# 13. SAVE CONVERSATION  
# =====  
conversation_history.append(  
    (question, answer)  
)  
  
# 14. DISPLAY ANSWER  
# =====  
print("\nChatbot:", answer)  
  
print("\nProgram completed.")
```

Output (Right Window):

```
File Edit Shell Debug Options Window Help  
Loading embedding model...  
Embedding model loaded.  
  
Creating document embeddings...  
Embeddings created.  
FAISS vector database created.  
Vectors stored: 6  
  
Loading language model...  
Device set to use cpu  
Language model loaded.  
  
===== CHATBOT READY =====  
Ask questions about Artificial Intelligence.  
The chatbot remembers the previous conversation.  
Type 'exit' to stop.  
  
You: What is machine learning?  
Chatbot: Deep Learning is a subset of Artificial Intelligence that allows computers to learn from data.  
  
You: What is its main purpose?  
Chatbot: Reinforcement Learning is a subset of Artificial Intelligence that allows computers to learn from data.  
  
You: What is deep learning?  
Chatbot: Deep Learning is a type of Machine Learning that uses neural networks with multiple layers.  
  
You: exit  
Chatbot: Goodbye!  
  
Program completed.  
>>>
```

